



INFINI TELCO

Customer Segmentation & Product Category Analysis

GROUP 7

Final Capstone Presentation



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EXECUTIVE SUMMARY

Project Overview

This project presents an end-to-end data analytics and customer intelligence initiative for Infini Telco, a major B2B telecommunications distributor in Malaysia. Utilizing a rich dataset spanning 2.5 years (June 2020 to December 2022) consisting of over 518K transactions from 4,010 corporate customers, this project deploys advanced descriptive analytics and K-Means clustering to uncover granular customer behaviors and optimize the company's product portfolio.

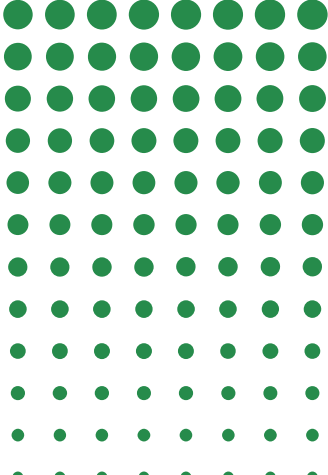




BUSINESS UNDERSTANDING

Infini Telco functions as a B2B telecommunications distributor within Malaysia's expanding digital economy. Because B2B sales cycles are inherently longer, involve larger deal sizes, and depend heavily on recurring relationship-based contracts, understanding exact account behaviors is vital.





BUSINESS OBJECTIVES AND PROBLEM STATEMENT

Infini Telco operates in a highly mature and competitive B2B telco landscape. The primary business challenges include a lack of clear visibility into high-value customer accounts, unoptimized cross-selling strategies, and an uneven distribution of revenue across its product categories.

The core objectives of this project are:

- Segment the customer base using behavioral metrics (revenue, transaction frequency, and invoice volumes) to separate high-value enterprise accounts from low-activity groups.
- Evaluate the product portfolio to identify key revenue drivers and underperforming categories.
- Formulate data-driven strategies to maximize customer retention, expand average revenue per customer, and optimize marketing spend.



BUSINESS QUESTION

BQ1: How can Infini Telco segment its corporate customers based on purchasing behavior, transaction activity, and revenue contribution?

BQ2: Which customer segments and product categories contribute the most to overall business performance and revenue generation?

BQ3: How can customer and product purchasing patterns be leveraged to optimize marketing strategies, cross-selling opportunities, and resource allocation?

SCOPE OF ANALYSIS

The analysis evaluates 2.5 years of historical B2B transactional data, cross-referencing customer demographic profiles, account mapping structures, and a multi-tiered product hierarchy. The technical scope includes:

1. Data Engineering & Preparation: Cleaning join keys, removing type mismatches, and handling data irregularities.
2. Advanced Analytics: K-Means Clustering for behavioral customer segmentation.
 - Pareto Analysis (80/20 Rule) to determine the high-impact product categories driving the business.
3. Dashboard & Visualization Development: Creating interactive Power BI dashboards focused on Customer Segmentation and Product Categorization.

Strategic Pillar	Focus Area	Expected Business Impact
Customer Retention	Loyalty programs & proactive engagement for Cluster 3	↑ 15% – 25% improvement in customer retention and retained revenue stability.
Revenue Growth	Ecosystem product bundling and cross-selling to Cluster 4 & 0	Upsell pipeline expansion, increasing overall average revenue per customer from its baseline of RM232.18K.
Product Optimization	Reallocating capital from low-performing long-tail items to core assets	Enhanced inventory efficiency, reduced operational overhead, and specialized marketing spend efficiency.



BUSINESS CHALLENGES

1. Customer Acquisition and Retention Challenges

Acquiring corporate clients requires significant upfront marketing and sales investments. However, without a granular view of customer lifetime value, Infini Telco runs the risk of over-allocating account management resources to low-margin groups while neglecting high-value enterprise players. Retaining a customer base where Cluster 3 alone commands over half of the company's revenue requires precise account health monitoring. Losing even a small percentage of this group would trigger severe financial destabilization.

2. Product Adoption and Cross-Selling Opportunities

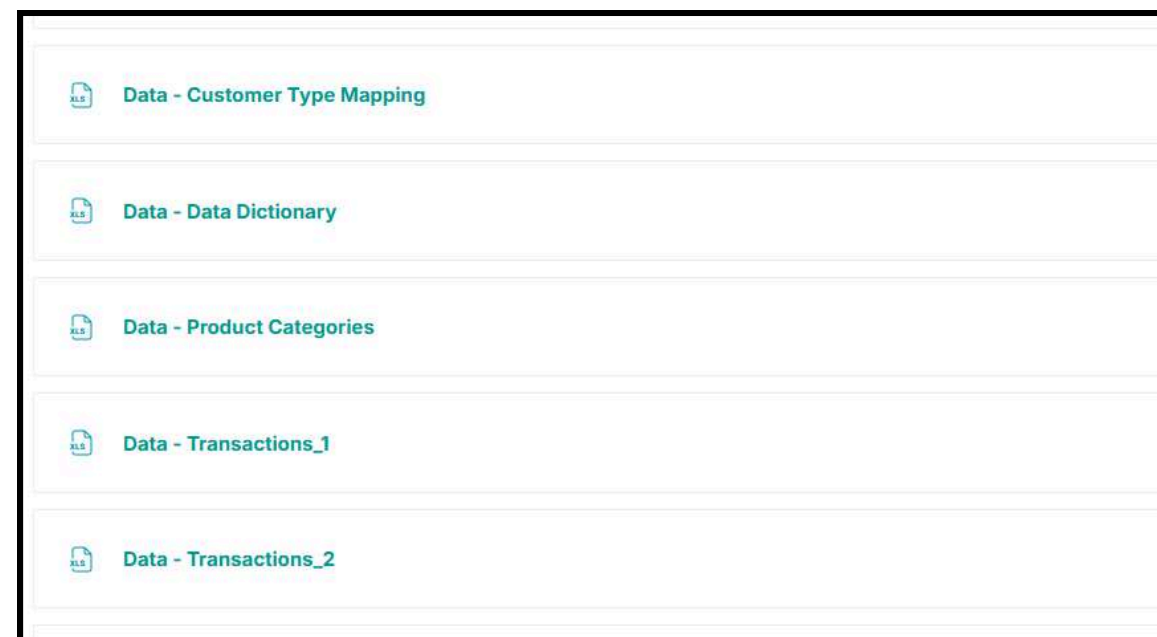
Infini Telco boasts a diverse portfolio of 21 categories, yet customer adoption is deeply lopsided. Over 64% of clients limit their engagement strictly to "Core" product lines. The organization faces a clear barrier in motivating customers to cross over into Standard, Specialty, or Long-tail categories. This represents an untapped goldmine: the company already possesses the billing relationships, but lacks the strategic coordination needed to transition single-category buyers into multi-service ecosystem adopters.



DATA UNDERSTANDING & PREPARATION

OVERVIEW OF THE DATASETS

Infini Telco is a B2B telco distributor in Malaysia. The raw data covers transactions from June 2020 to December 2022 which is roughly 2.5 years of sales records. The datasets consists of 3 transaction files, with another 3 supplementary mapping files that describe who the customers are and what the products are.



Transactions_1/2/3.csv	Raw sales records such as invoice, customer, product, turnover, amount
Customer_Characteristics.csv	Customer name, account group, and company status
Customer_Type_Mapping.csv	Maps Main Account ID to a customer type
Product_Categories.csv	Maps Product ID to ProdCat2 (granular) and ProdCat1 (broad)

DATA PREPARATION

Combining the 3 Transaction Files and Validating the Values

```
df1 = pd.read_csv('Transactions_1.csv')
df2 = pd.read_csv('Transactions_2.csv')
df3 = pd.read_csv('Transactions_3.csv')

# Combine all dataframes into one dataframe
df_Txn_full = pd.concat([df1, df2, df3], ignore_index=True)
```

```
# Validate Gross Annual Turnover
print("Gross Annual Turnover:")
print(df_Txn_full['GROSS_TURNOVER'].sum())

# Validate total unique products
print("\nTotal Unique Products:")
print(df_Txn_full['PRODUCT'].nunique())

# Validate total customer base
print("\nTotal Unique Customers:")
print(df_Txn_full['CUSTOMER_NUMBER'].nunique())
```

```
Gross Annual Turnover:
1466478196.7234998
```

```
Total Unique Products:
191394
```

```
Total Unique Customers:
7130
```

DATA PREPARATION

Cleaning the Join Keys

```
# Clean customer_number column
df_Txn_full['CUSTOMER_NUMBER'] = (
    df_Txn_full['CUSTOMER_NUMBER']
    .astype(str)
    .str.strip()
    .str.lstrip('0')
)

# Clean Product column
df_Txn_full['PRODUCT'] = (
    df_Txn_full['PRODUCT']
    .astype(str)
    .str.strip()
    .str.lstrip('0')
)

# Clean Main_account column
df_Txn_full['Main_account'] = (
    df_Txn_full['Main_account']
    .astype(str)
    .str.strip()
    .str.lstrip('0')
)

# Clean Invoice_Number column
df_Txn_full['INVOICE_NUMBER'] = (
    df_Txn_full['INVOICE_NUMBER']
    .astype(str)
    .str.strip()
    .str.lstrip('0')
)
```

.astype(str)	Forces everything to text, avoids type mismatch (int vs string)
.str.strip()	Removes accidental spaces before/after the value
.str.lstrip('0')	Removes leading zeroes, so '00123' becomes '123'

DATA PREPARATION

Joining the 3 Mapping Files

```
# Clean Customer_Number column in df_CustChar
df_CustChar['CUSTOMER_NUMBER'] = (
    df_CustChar['CUSTOMER_NUMBER']
    .astype(str)
    .str.strip()
    .str.lstrip('0')
)

# Remove duplicates
df_Txn_full = df_Txn_full.drop_duplicates()
df_CustChar = df_CustChar.drop_duplicates()

# Create master dataframe
df_Master = pd.merge(
    df_Txn_full,
    df_CustChar,
    left_on='CUSTOMER_NUMBER',
    right_on='CUSTOMER_NUMBER',
    how='left'
)

# Display master dataframe
df_Master.head()
```

```
# Drop columns from df_Master that would be duplicated or overwritten by df_CustType
# This handles cases where the cell is run multiple times
columns_to_drop = ['Main_account_ID', 'Customer_type']
for col in columns_to_drop:
    if col in df_Master.columns:
        df_Master = df_Master.drop(columns=[col])

# Join df_CustType with df_Master
df_Master = pd.merge(
    df_Master,
    df_CustType,
    left_on='Main_account',
    right_on='Main_account_ID',
    how='left'
)

# Display dataframe
df_Master.head()
```

```
# Clean ProductID column
df_ProdCategories['ProductID'] = (
    df_ProdCategories['ProductID']
    .astype(str)
    .str.strip()
    .str.lstrip('0')
)

# Remove duplicates
df_ProdCategories = df_ProdCategories.drop_duplicates()

# Join df_ProdCategories with df_Master
df_Master = pd.merge(
    df_Master,
    df_ProdCategories,
    left_on='PRODUCT',
    right_on='ProductID',
    how='left'
)

# Display dataframe
df_Master.head()
```


DATA PREPARATION

Creating Invoice & SKU Flags

Task 1.3.1: In the df_Master data, add a column Invoice_flag, reflecting 0 if number of transactions <= 3, otherwise 1

This is to flag customers who purchase less frequently

```
# Count transactions for each customer
invoice_count = df_Master.groupby('CUSTOMER_NUMBER')['INVOICE_NUMBER'].nunique()

# Create Invoice_flag
df_Master['Invoice_flag'] = df_Master['CUSTOMER_NUMBER'].map(
    lambda x: 0 if invoice_count[x] <= 3 else 1
)

# Display sample
df_Master[['CUSTOMER_NUMBER', 'Invoice_flag']].head()
```

Task 1.3.2: In the df_Master data, add a column Invoice_flag, reflecting 0 if number of transactions <= 3, otherwise 1

This is to flag customers who purchase single SKU.

```
# Count unique SKU purchased by each customer
sku_count = df_Master.groupby('CUSTOMER_NUMBER')['PRODUCT'].nunique()

# Create SKU_flag
df_Master['SKU_flag'] = df_Master['CUSTOMER_NUMBER'].map(
    lambda x: 0 if sku_count[x] <= 1 else 1
)

# Display sample
df_Master[['CUSTOMER_NUMBER', 'SKU_flag']].head()
```

	CUSTOMER_NUMBER	SKU_flag
0	14816	1

DATA PREPARATION

Data Cleaning

Task 1.3.3: In the master data, remove records where Product ID is null and save it in a dataframe df_Filtered

```
# Remove records where ProductID is null
df_Filtered = df_Master[df_Master['ProductID'].notnull()]
```

Task 1.3.4: In the df_Filtered dataframe, remove records where Product Category information is null

```
# Remove records where Product Category is null
df_Filtered = df_Filtered[df_Filtered['ProdCat2'].notnull()]
```

Task 1.3.5: In the df_Filtered dataframe, remove records pertaining to sales to Internal Account

Hint: Check for Internal accounts in Account_Group_TXT field

```
# Remove Internal Account records
df_Filtered = df_Filtered[
    df_Filtered['ACCOUNT_GROUP_TXT'] != 'Internal Account'
]
```

Task 1.3.6: In the df_Filtered dataframe, remove records with position types "ZSKA" and "ZDIR"

Hint: Check column Position_type_order_line

```
# Remove ZSKA and ZDIR position types
df_Filtered = df_Filtered[
    ~df_Filtered['Position_type_order_line'].isin(['ZSKA', 'ZDIR'])
]
```

DATA PREPARATION

Data Cleaning

Task 1.3.7: In the df_Filtered dataframe, remove records whose company status are not active/normal

Hint: Check column CVR_Company_status

```
# Keep only active/normal companies OR null values
df_Filtered = df_Filtered[
    (df_Filtered['CVR_Company_status'].isin(['Active', 'Normal'])) |
    (df_Filtered['CVR_Company_status'].isnull())
]
```

Task 1.3.8: In the df_Filtered dataframe, remove records where Gross Turnover is zero or negative

```
# Remove records where Gross Turnover <= 0
df_Filtered = df_Filtered[df_Filtered['GROSS_TURNOVER'] > 0]
```

Task 1.3.9: In the df_Filtered dataframe, remove records where Amount is zero or negative

```
# Remove records where Amount <= 0
df_Filtered = df_Filtered[df_Filtered['AMOUNT'] > 0]
```

Task 1.3.10: In the df_Filtered dataframe, filter out customers who have made 3 or fewer transactions

Hint: Utilize the Invoice_flag column created earlier.

```
# Filter customers with more than 3 transactions
df_Filtered = df_Filtered[df_Filtered['Invoice_flag'] == 1]
```


DATA PREPARATION

Data Cleaning

Task 1.3.11: In the df_Filtered dataframe, filter out customers who have purchased only a single SKU

Hint: Utilize the SKU_flag column created earlier.

```
# Filter customers who purchased more than one SKU
df_Filtered = df_Filtered[df_Filtered['SKU_flag'] == 1]
```

Task 1.3.12: In the df_Filtered dataframe, Filter out "Key" customer accounts

Hint: Check Customer_type column

```
# Remove Key customer accounts
df_Filtered = df_Filtered[
    ~df_Filtered['Customer_type'].str.contains('Key', case=False, na=False)
]
```

DATA PREPARATION

Pareto Analysis (80/20 Rule)

Identify the top product categories that drive 80% of revenue

Task 2.2: Create a new dataframe `df_Pareto` with the columns: `ProdCat2`, `Gross_Turnover`, `Cumulative_GT`, `Cumulative_GT%`

Hint: Cumulative_GT is Cumulative Gross Turnover which you can get by doing a cumulative sum on the Gross Turnover.

```
df_Pareto = df_Filtered.groupby('ProdCat2')['GROSS_TURNOVER'].sum().reset_index()

df_Pareto = df_Pareto.sort_values(by='GROSS_TURNOVER', ascending=False)

df_Pareto['Cumulative_GT'] = df_Pareto['GROSS_TURNOVER'].cumsum()

df_Pareto['Cumulative_GT%'] = (
    df_Pareto['Cumulative_GT'] /
    df_Pareto['GROSS_TURNOVER'].sum()
) * 100
```

Task 2.3: Arrive at a "Pareto filtered" dataset `df_Pareto_80` with the top product categories that contribute to 80% of the Gross Turnover

Hint: There are 64 products that contribute to 80% of Gross Revenue turnover

```
df_Pareto_80 = df_Pareto[df_Pareto['Cumulative_GT%'] <= 80]

print(df_Pareto_80.shape)

(65, 4)
```

DATA PREPARATION

Pareto Analysis (80/20 Rule)

Identify the top product categories that drive 80% of revenue

Task 2.2: Create a new dataframe `df_Pareto` with the columns: `ProdCat2`, `Gross_Turnover`, `Cumulative_GT`, `Cumulative_GT%`

Hint: Cumulative_GT is Cumulative Gross Turnover which you can get by doing a cumulative sum on the Gross Turnover.

```
df_Pareto = df_Filtered.groupby('ProdCat2')['GROSS_TURNOVER'].sum().reset_index()

df_Pareto = df_Pareto.sort_values(by='GROSS_TURNOVER', ascending=False)

df_Pareto['Cumulative_GT'] = df_Pareto['GROSS_TURNOVER'].cumsum()

df_Pareto['Cumulative_GT%'] = (
    df_Pareto['Cumulative_GT'] /
    df_Pareto['GROSS_TURNOVER'].sum()
) * 100
```

Task 2.3: Arrive at a "Pareto filtered" dataset `df_Pareto_80` with the top product categories that contribute to 80% of the Gross Turnover

Hint: There are 64 products that contribute to 80% of Gross Revenue turnover

```
df_Pareto_80 = df_Pareto[df_Pareto['Cumulative_GT%'] <= 80]

print(df_Pareto_80.shape)

(65, 4)
```

Task 2.5: Remove the redundant columns and give a meaningful name to the repetitive columns

```
df_Master_Pareto = df_Master_Pareto.loc[:, ~df_Master_Pareto.columns.duplicated()]
```


DATA PREPARATION

Customer Segmentation

Preparing Data for K-Means

```
df_segment = df_Master_Pareto.copy()

# Create primary key

df_segment['Primary_Key'] = (
    df_segment['Main_account'].astype(str) + ' - ' +
    df_segment['Main_account_name'].astype(str)
)

# Group turnover

df_segment = df_segment.groupby(
    ['Primary_Key', 'ProdCat1']
)['GROSS_TURNOVER'].sum().reset_index()

# Calculate spend %

total_spend = df_segment.groupby('Primary_Key')['GROSS_TURNOVER'].transform('sum')

df_segment['Customer Spend %'] = (
    df_segment['GROSS_TURNOVER'] / total_spend
) * 100
```

K-Means needs one row per customer, with numeric columns. We need to transform the millions of transaction rows into, for each customer, what % of their total spend went to each product category (ProdCat1). This spend % matrix is what the algorithm uses to find similar customers.

Task 3.2: Pivot the dataset to create a matrix view of spend % of all customers on all products (ProdCat1). Save the dataframe as df_pivot. Create another dataframe df_clustering removing the primary key from the df_pivot dataframe.

```
df_pivot = df_segment.pivot_table(
    index='Primary_Key',
    columns='ProdCat1',
    values='Customer Spend %',
    fill_value=0
).reset_index()

# Remove primary key for clustering

df_clustering = df_pivot.drop('Primary_Key', axis=1)
```

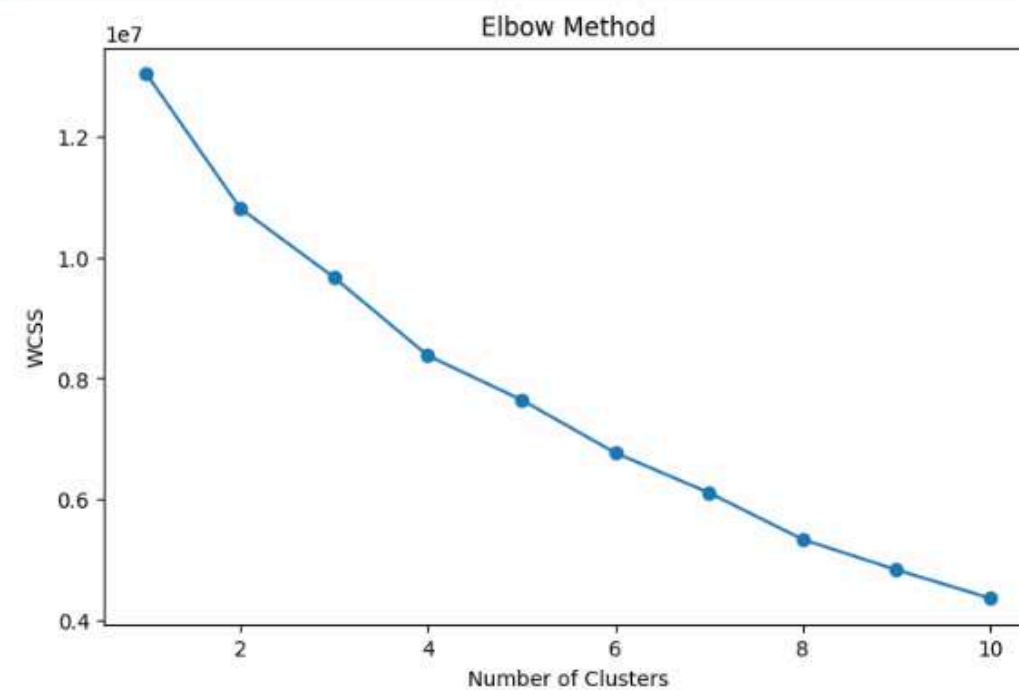
DATA PREPARATION

Customer Segmentation

K-Means Clustering

Task 3.3: Run K-Means clustering algorithm and identify the optimal number of clusters using Elbow Method

```
wcss = []  
  
for i in range(1, 11):  
    kmeans = KMeans(n_clusters=i, random_state=42)  
    kmeans.fit(df_clustering)  
    wcss.append(kmeans.inertia_)  
  
plt.figure(figsize=(8,5))  
plt.plot(range(1,11), wcss, marker='o')  
plt.xlabel('Number of Clusters')  
plt.ylabel('WCSS')  
plt.title('Elbow Method')  
plt.show()
```



The Elbow Curve provides an optimal value for 'k' that is the number of clusters to be generated in the output. This value can be used as a guideline for the actual clustering run; one may vary the number of clusters in the output based on business need.

Task 3.4: Run the KMeans algorithm with optimal number of clusters

```
kmeans = KMeans(n_clusters=10, random_state=42)  
clusters = kmeans.fit_predict(df_clustering)
```

Task 3.5: From the above clusters, create cluster profile

Task 3.5.1: Map each customer to their respective cluster in df_segment dataset

```
df_pivot['Cluster'] = clusters  
  
# Merge back  
  
df_segment = pd.merge(  
    df_segment,  
    df_pivot[['Primary_Key', 'Cluster']],  
    on='Primary_Key',  
    how='left'  
)
```

- Each customer gets a Cluster label (0–9)
- Customers in the same cluster have similar proportional spending across product categories

DATA PREPARATION

Prepare data for Dashboard

```
# — Step 1: Add Primary_Key and Cluster to df_Master_Pareto —————
df_Master_Pareto['Primary_Key'] = (
    df_Master_Pareto['Main_account'].astype(str) + ' - ' +
    df_Master_Pareto['Main_account_name'].astype(str)
)

if 'Cluster' not in df_Master_Pareto.columns:
    df_Master_Pareto = pd.merge(
        df_Master_Pareto,
        df_pivot[['Primary_Key', 'Cluster']],
        on='Primary_Key',
        how='left'
    )

# — Step 2: Build the full ProdCat1 x ProdCat2 skeleton (219 combos) —————
all_prodcats = df_Master_Pareto[['ProdCat1', 'ProdCat2']].drop_duplicates()

# — Step 3: Get all unique customers with their cluster —————
all_customers = (
    df_Master_Pareto[['Main_account', 'Main_account_name', 'Cluster']]
    .drop_duplicates()
)
all_customers['Main_account_name_number'] = (
    all_customers['Main_account'].astype(str)
    + '_'
    + all_customers['Main_account_name'].astype(str)
)

# — Step 4: Cross join customers with all product categories —————
all_customers['_key'] = 1
all_prodcats['_key'] = 1
df_skeleton = pd.merge(all_customers, all_prodcats, on='_key').drop('_key', axis=1)

# — Step 5: Aggregate actual transactions —————
df_actual = (
    df_Master_Pareto
    .groupby(['Main_account', 'Main_account_name', 'ProdCat1', 'ProdCat2'])
    .agg(
        GROSS_TURNOVER=('GROSS_TURNOVER', 'sum'),
        INVOICE_NUMBER=('INVOICE_NUMBER', 'nunique')
    )
    .reset_index()
)
```

```
# — Step 6: Left join skeleton with actuals (fills 0 for unvisited combos) —————
df_tableau = pd.merge(
    df_skeleton,
    df_actual,
    on=['Main_account', 'Main_account_name', 'ProdCat1', 'ProdCat2'],
    how='left'
)
df_tableau['GROSS_TURNOVER'] = df_tableau['GROSS_TURNOVER'].fillna(0)
df_tableau['INVOICE_NUMBER'] = df_tableau['INVOICE_NUMBER'].fillna(0).astype(int)

# — Step 7: Keep only required columns in the right order —————
df_tableau = df_tableau[[
    'Cluster', 'ProdCat1', 'ProdCat2',
    'GROSS_TURNOVER', 'INVOICE_NUMBER',
    'Main_account_name_number'
]]

print(df_tableau.shape)
print(df_tableau.head())
```

```
(269945, 6)
   Cluster  ProdCat1  ProdCat2 \
0         1  Digital Transformation Services  Cloud Strategy and Migration
1         1         Data and Analytics  Predictive Analytics
2         1  Television and Entertainment  Mobile TV Services
3         1  Broadband and Internet Services  4G Services
4         1  Digital Transformation Services  Cloud and Hosting Services

   GROSS_TURNOVER  INVOICE_NUMBER  Main_account_name_number
0         310992.76             174  103350_Customer Group_24
1         484361.08             312  103350_Customer Group_24
2         158225.14             184  103350_Customer Group_24
3          89662.06              61  103350_Customer Group_24
4         128353.05              47  103350_Customer Group_24
```


DATA PREPARATION

Scatter Plot: Gross Turnover vs Invoice Count

```
# — Scatter plot: Gross Turnover vs Invoice Count per customer, coloured by cluster —

# Aggregate to one row per Main Account
df_scatter = df_Master_Pareto.groupby(['Main_account', 'Main_account_name', 'Cluster']).agg(
    Gross_Turnover=('GROSS_TURNOVER', 'sum'),
    Invoice_Count=('INVOICE_NUMBER', 'nunique')
).reset_index()

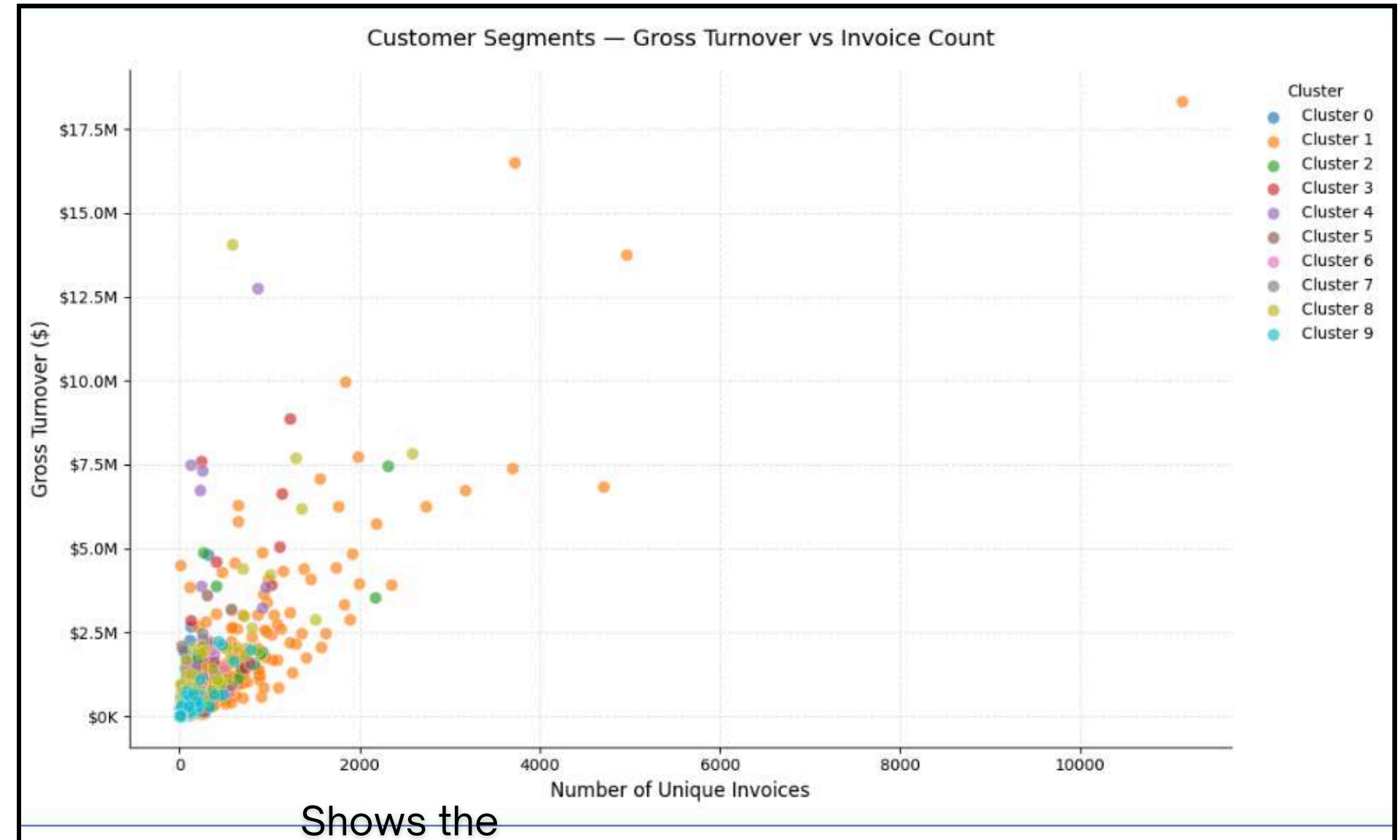
# Colour palette — one colour per cluster
n_clusters = df_scatter['Cluster'].nunique()
palette = sns.color_palette('tab10', n_colors=n_clusters)
color_map = {cluster: palette[i] for i, cluster in enumerate(sorted(df_scatter['Cluster'].unique()))}

fig, ax = plt.subplots(figsize=(12, 7))

for cluster_id, group in df_scatter.groupby('Cluster'):
    ax.scatter(
        group['Invoice_Count'],
        group['Gross_Turnover'],
        label=f'Cluster {cluster_id}',
        color=color_map[cluster_id],
        alpha=0.65,
        s=60,
        edgecolors='white',
        linewidths=0.4
    )

# Formatting
ax.set_xlabel('Number of Unique Invoices', fontsize=12)
ax.set_ylabel('Gross Turnover ($)', fontsize=12)
ax.set_title('Customer Segments — Gross Turnover vs Invoice Count', fontsize=14, pad=15)
ax.yaxis.set_major_formatter(FuncFormatter(lambda x, _: f'${x/1e6:.1f}M' if x >= 1e6 else f'${x/1e3:.0f}K'))
ax.legend(title='Cluster', bbox_to_anchor=(1.01, 1), loc='upper left', frameon=False)
ax.spines[['top', 'right']].set_visible(False)
ax.grid(axis='both', linestyle='--', alpha=0.3)

plt.tight_layout()
plt.show()
```



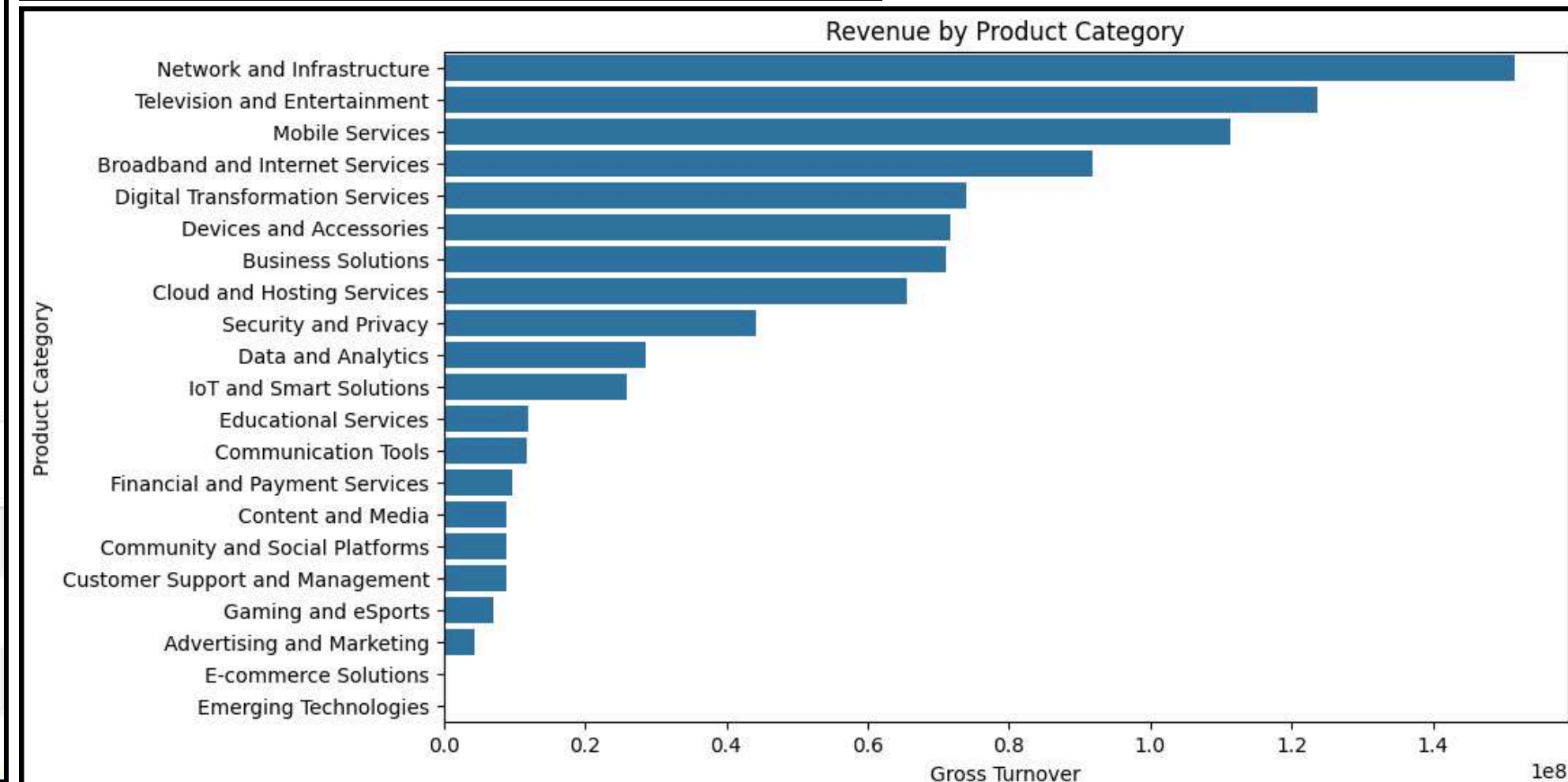
DATA PREPARATION

Product Categorization

```
def classify_product(x):  
    if x <= 70:  
        return 'Core'  
  
    elif x <= 90:  
        return 'Standard'  
  
    elif x <= 98:  
        return 'Specialty'  
  
    else:  
        return 'Long-tail'  
  
product_sales['Product_Type'] = product_sales[  
    'Cum_Percentage'  
].apply(classify_product)  
  
product_sales.head()
```

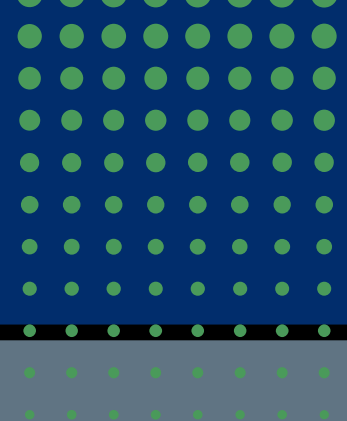
...	ProdCat1	GROSS_TURNOVER	Cum_Percentage	Product_Type
18	Network and Infrastructure	1.514958e+08	16.271580	Core
20	Television and Entertainment	1.236869e+08	29.556314	Core
17	Mobile Services	1.112904e+08	41.509585	Core
1	Broadband and Internet Services	9.179030e+07	51.368428	Core

```
plt.figure(figsize=(10,6))  
  
sns.barplot(  
    data=product_sales,  
    x='GROSS_TURNOVER',  
    y='ProdCat1'  
)  
  
plt.title('Revenue by Product Category')  
plt.xlabel('Gross Turnover')  
plt.ylabel('Product Category')  
plt.show()
```





ANALYTICS METHODOLOGY



Customer Segmentation Methodology



PRODUCT CATEGORIZATION METHODOLOGY



Model Selection Rationale

Method Reason Selected Customer Segmentation

K-Means Clustering

Suitable for customer segmentation. Groups customers with similar purchasing behaviors for effective customer segmentation.

-The Elbow Method in KMeans Clustering

The Elbow Method is a commonly used technique to determine the optimal number of clusters (K) in K-Means clustering. It helps to balance between underfitting and overfitting by finding a point where adding more clusters doesn't significantly improve the model performance.

WCSS (Within Cluster Sum of Squares) vs. Number of Clusters:

-The x-axis represents the number of clusters (K).

The y-axis represents the Within-Cluster Sum of Squares (WCSS), which measures the variance within each cluster.

-The point marked on the plot indicates the "elbow point" where the rate of decrease in WCSS slows down.

-Before the Elbow Point: Adding more clusters significantly reduces WCSS, indicating that the clusters are becoming more defined and compact.

After the Elbow Point: The reduction in WCSS slows down, meaning that adding more clusters doesn't significantly improve the clustering.

Model Selection Rationale

Method reason selected product categorization

Descriptive Analytics

- Easy interpretation of product performance.
- Summarizes historical data to understand product performance and business trends.
- Instead of predicting future outcomes, this method focuses on providing a clear overview of existing trends and patterns.

Evaluation Metrics & Validation Approach

Customer Segmentation Evaluation

- Cluster Separation:
- Total Revenue, Total Invoice, Average Revenue per Customer, Number of Customers
 - Cluster 3 contains 1,613 customers and generates approximately RM537.1 million revenue, making it the highest-value segment.
 - Cluster 5, 6, and 7 contribute significantly lower average revenue per customer, indicating lower-value customer groups.

- Average Revenue per Customer :
- The significant differences indicate that the clustering model effectively separates customers according to their value.
- Example:
- Cluster 3 = RM332,954 per customer
 - Cluster 4 = RM135,942 per customer
 - Cluster 6 = RM61,434 per customer

- Revenue Contribution Analysis:
- Cluster 3 contributes more than half of the total revenue, indicating the model successfully identified the company's most valuable customer group.

- Compared revenue, invoice volume, and customer count across clusters.
- Verified that each cluster exhibited unique behavioral characteristics.
- Assessed whether high-value customers were clearly distinguishable from low-value customers.

Product Categorization Evaluation

- Revenue by Product Category: classified products into four categories; Core, Standard, Specialty, Long-tail
- Top-performing categories include:
- Network & Infrastructure = RM151.5M
 - Television & Entertainment = RM123.7M
 - Mobile Services = RM111.3M
 - Broadband & Internet Services = RM91.8M

- Customer Distribution by Product Type:
- Results show:
- Core products account for approximately 64.7% of customer distribution.
 - Standard products account for approximately 26.5%.
 - Specialty and Long-tail products represent smaller market shares.

- Revenue per Product:
- Average Product Price = RM16.38
 - Revenue per Product Category = RM30.38M

- Examined revenue contribution for each product category.
- Analyzed customer adoption patterns.
- Confirmed that product categories aligned with business performance and strategic priorities.

Validation



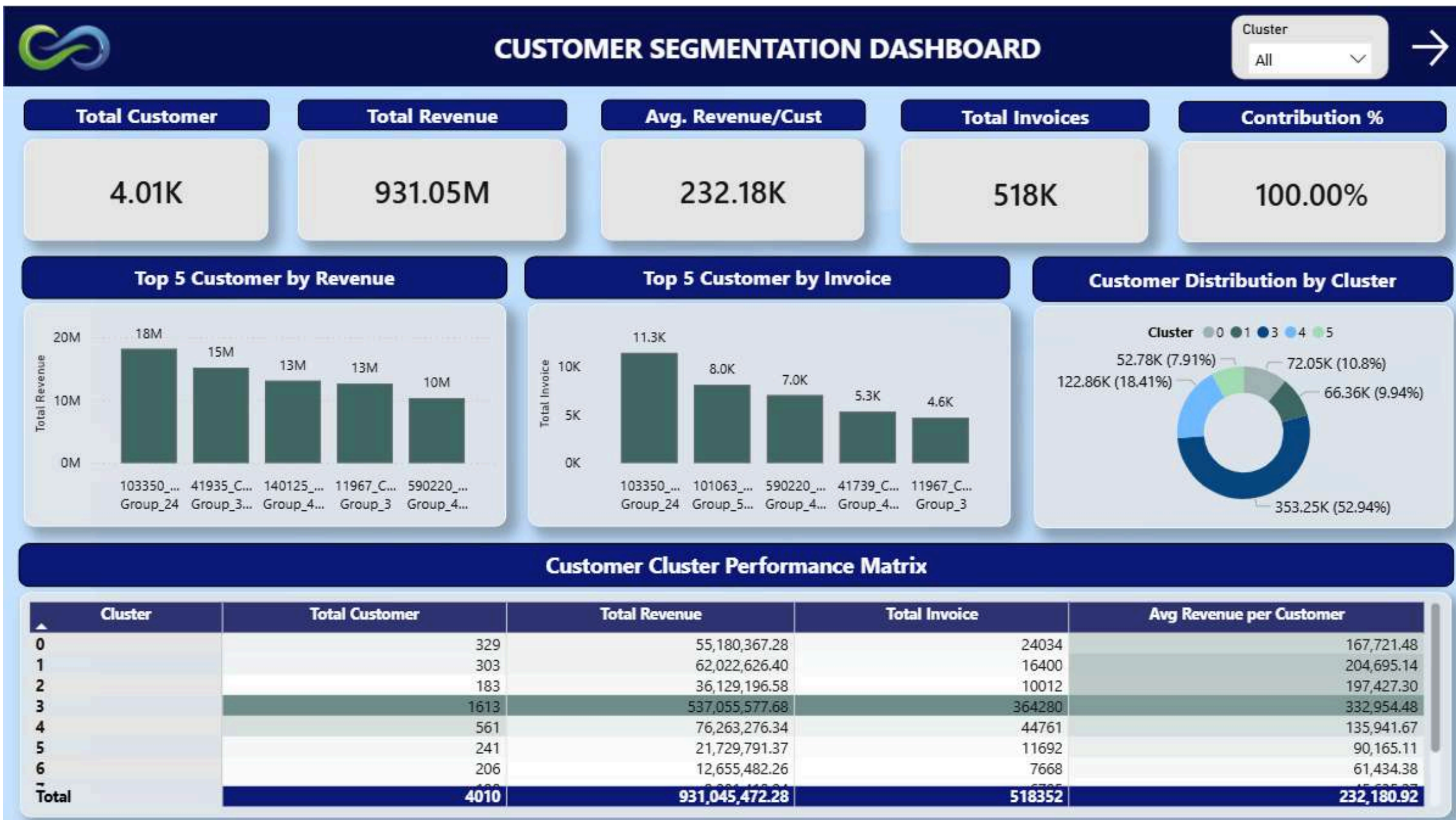
ANALYSIS RESULTS AND VISUALIZATION

CUSTOMER SEGMENTATION DASHBOARD



KEY BUSINESS INSIGHT

- Top 5 customers generate up to \$18M revenue individually
- Top invoice customers handle up to 11.3K transactions
- Customers with high invoices also generate high revenue
- High Value + High Activity → (priority customers)
- High Activity + Low Value → (upsell opportunity)
- Key account management and personalized engagement strategies.
- Maintaining strong relationships with top customers

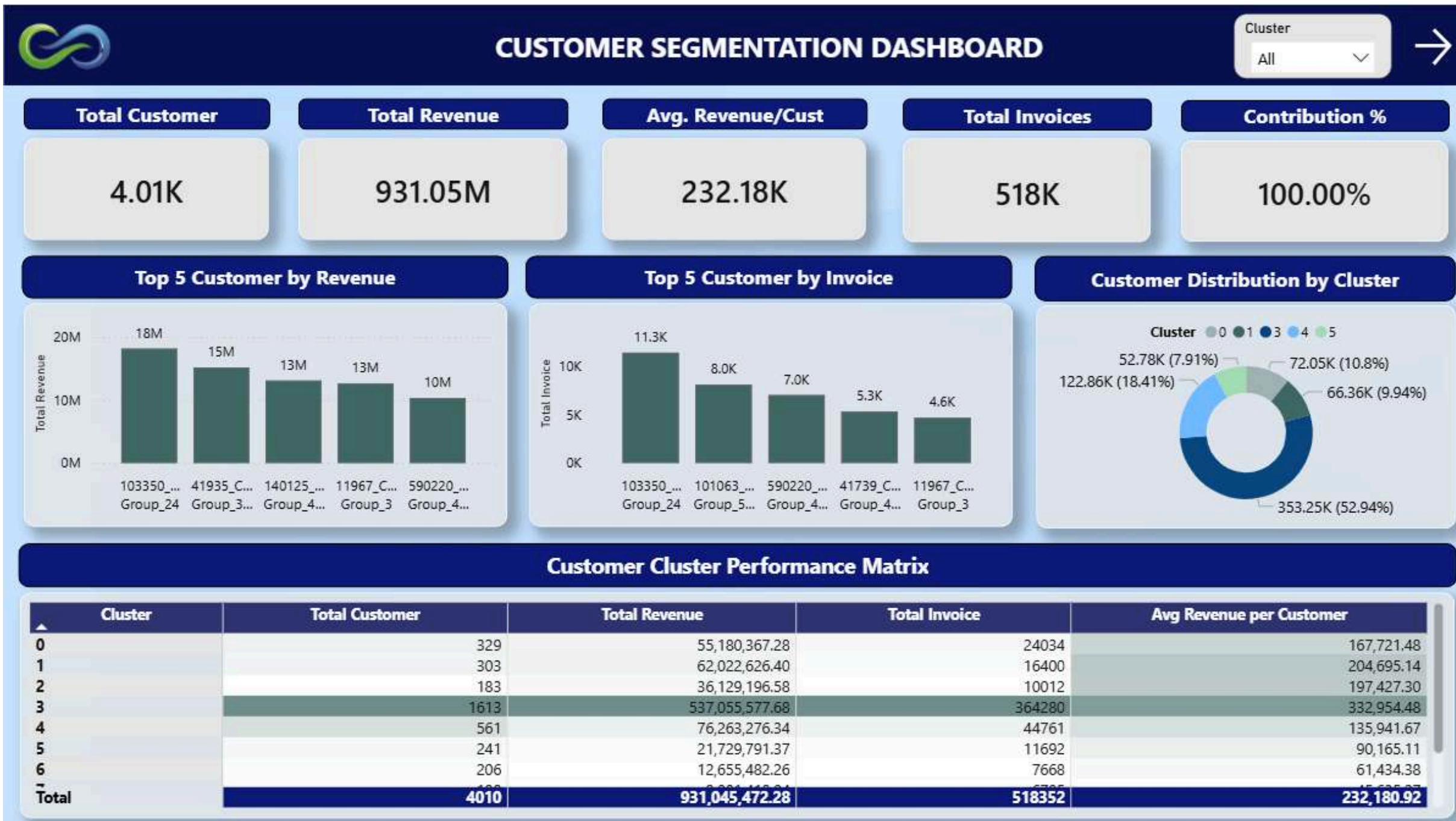


CUSTOMER SEGMENTATION DASHBOARD

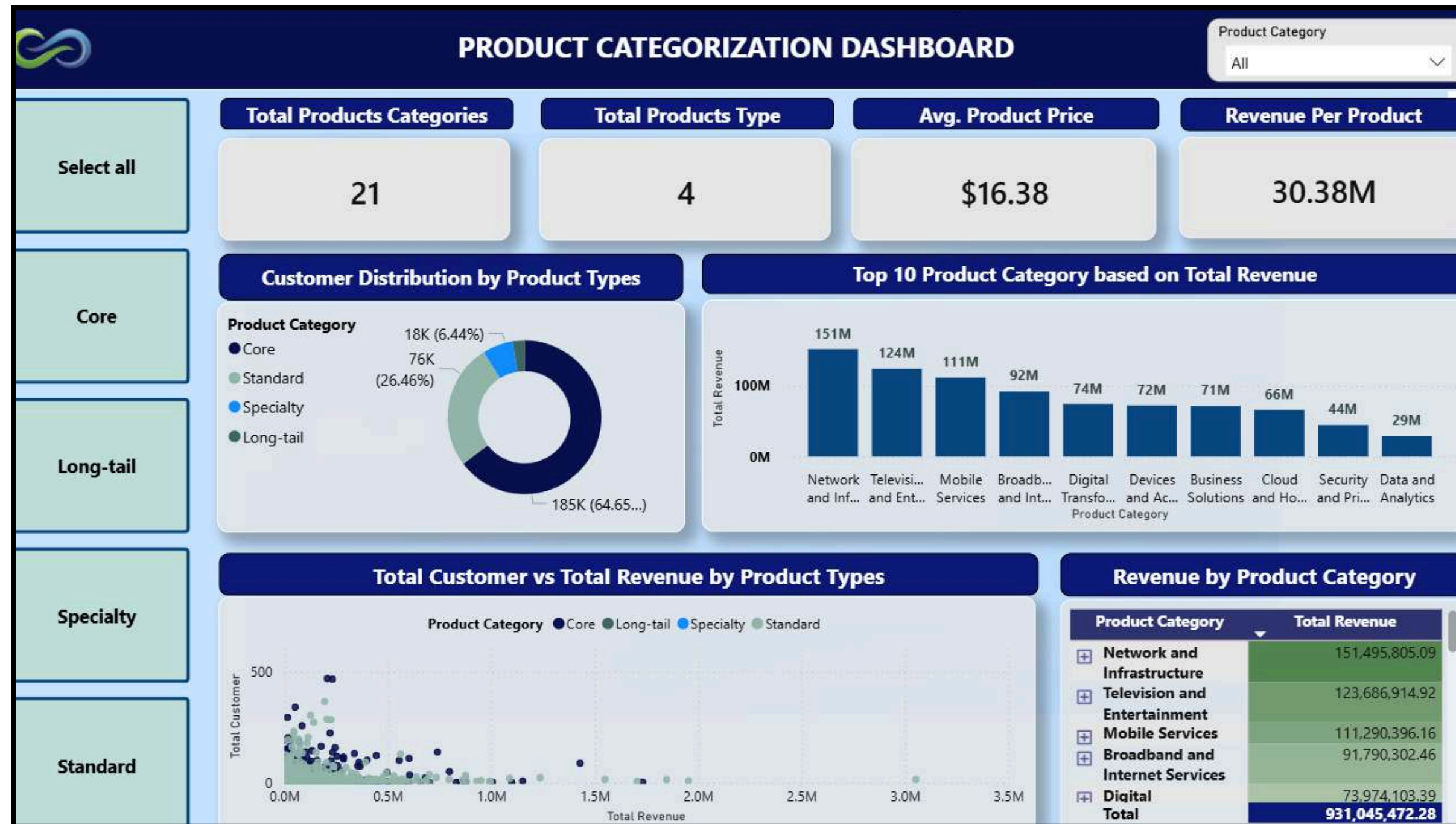


KEY BUSINESS INSIGHT

- Cluster 3 contributes ~52.94% of total customers ($\approx 353K$) and dominates revenue
- Cluster 3 has the highest average revenue per customer ($\sim \$332.95K$)
- Compared to overall average: \$232.18K
- Some clusters (Cluster 0, 1, 4) have customers but lower revenue contribution.
 - Cluster 4: 561 customers but lower avg revenue ($\sim \$135K$)
- Cluster 3 has the highest number of invoices (364,280)
- Strong link between activity (transactions) and revenue generation



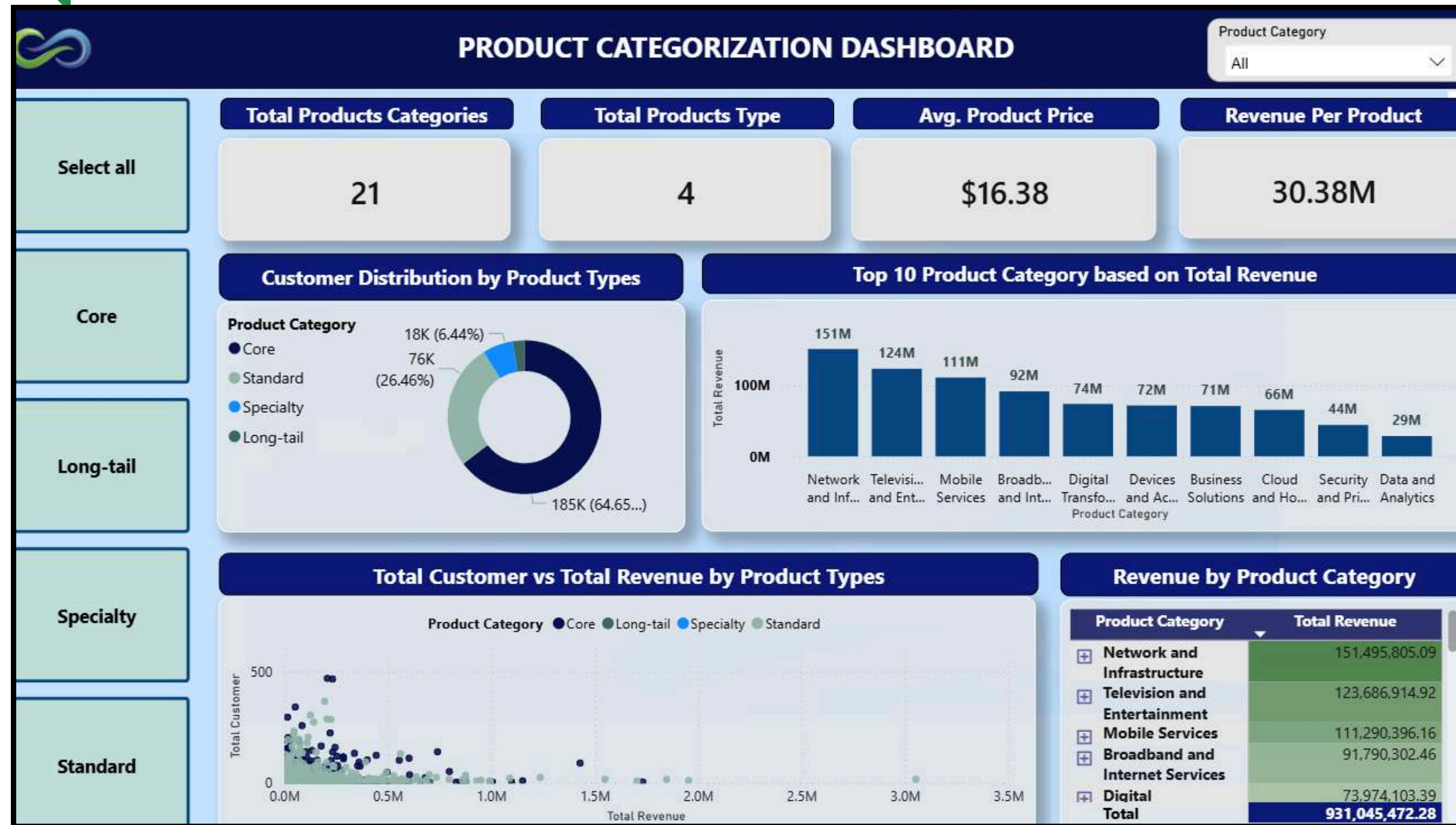
PRODUCT CATEGORIZATION DASHBOARD



Key Insights

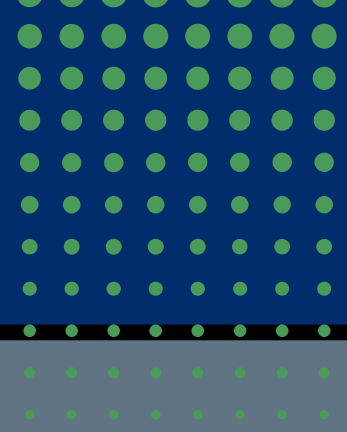
- ➔ Despite an **average product price** of only **\$16.38**, each product generates an **average revenue** of **\$30.38M**, indicating strong sales performance.
- ➔ The business has **21 product categories** grouped into only **4 product types**, making product management more efficient.
- ➔ Most products generate low revenue, while a few Standard products contribute over RM1 million in revenue.

PRODUCT CATEGORIZATION DASHBOARD



Key Insights

- ➔ The business is heavily reliant on the **Core** customer segment, which accounts for **over 64%** of the customer distribution.
- ➔ The **Data and Analytics** category appears as the **lowest performer** among the top 10, generating 29M in revenue compared to the 151M of the top-performing category.
- ➔ The **Network and Infrastructure** and **Television and Entertainment** categories are the primary revenue drivers for the company, contributing **151M** and **124M** respectively.



KEY INSIGHT



Largest Customer Segment

- Cluster 3 represents 52.94% of customers.
- More than half of customers belong to this segment.
- Retaining this segment is critical for business stability.

Customer Distribution by Cluster



Strong Revenue Generation

- Total revenue reached RM931.05 million.
- Average revenue per customer is RM232.18K.
- Existing customers contribute substantial business value.

Total Revenue

931.05M

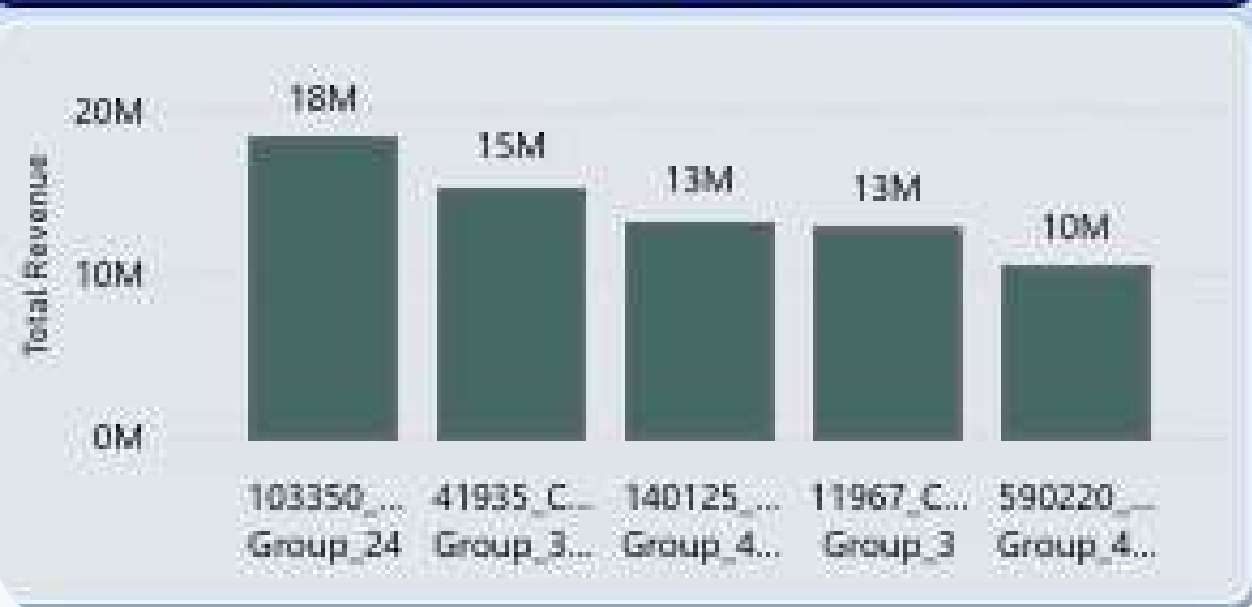
Ava. Revenue/Cust

232.18K

Revenue is Concentrated Among Key Customers Group

- The Top 5 customer groups by revenue contribute a disproportionately large share of total revenue. For example, Group 24 generated about RM18 M and Group 334 about RM15 M.
- Revenue performance depends heavily on a small number of high-value customer groups, so retaining and expanding these accounts should be a priority.

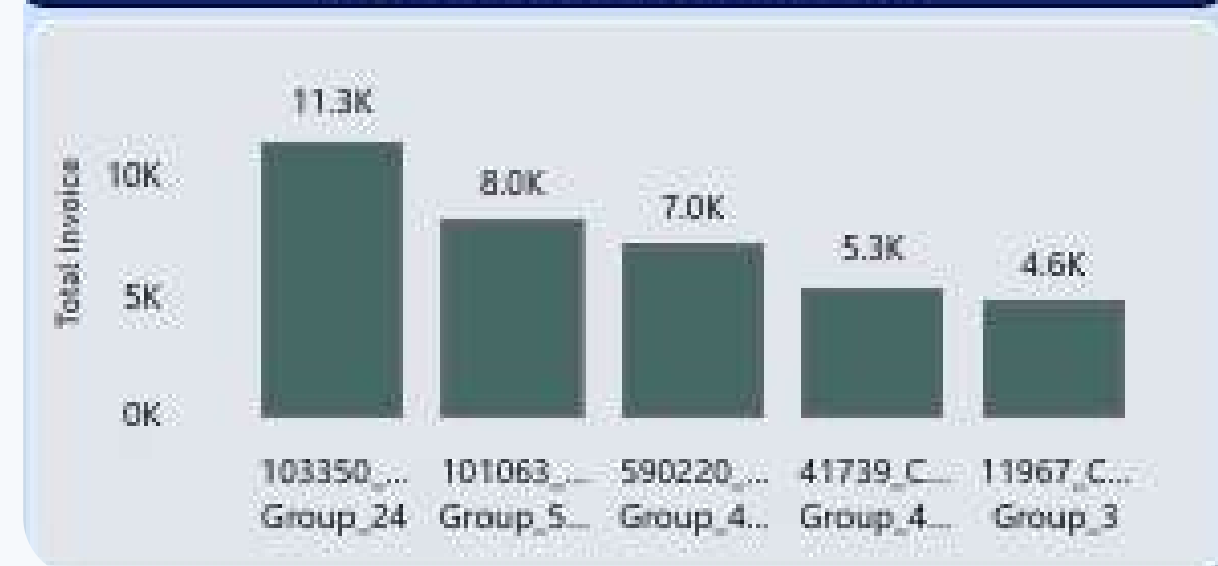
Top 5 Customer by Revenue



Certain Customers Groups Generate High Transaction Activity

- The Top 5 customer groups by invoice count account for a large volume of transactions and repeated purchases.
- These groups are highly engaged and are strong candidates for upselling, cross-selling, and long-term account development.

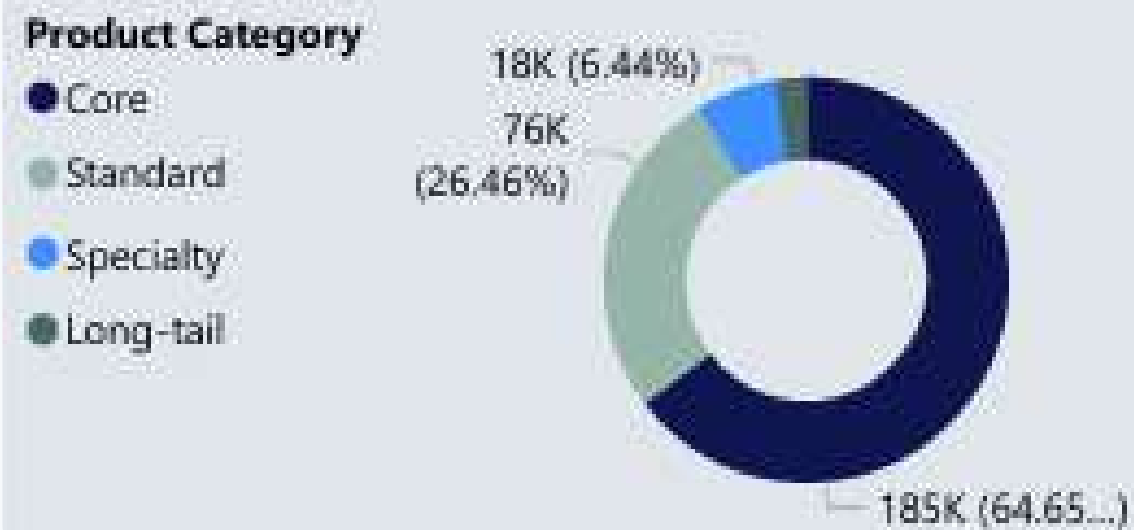
Top 5 Customer by Invoice



Core Products Dominate Customer Adoption

- From the product type distribution:
- Core $\approx 64\%$
- Long-tail $\approx 26\%$
- Specialty $\approx 6\%$
- Standard $\approx 4\%$
- Customers are primarily concentrated around Core products.

Customer Distribution by Product Types

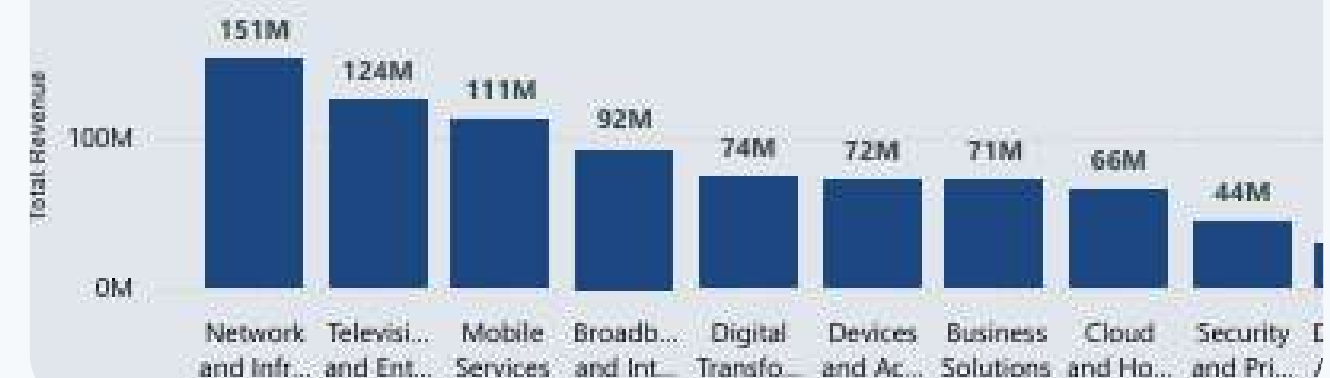


Product Portfolio Revenue is Uneven

- The Top 10 categories contribute the majority of revenue.
- Several categories contribute significantly less revenue.

This indicates dependency on a small number of successful products.

Top 10 Product Category based on Total Revenue





STRATEGIC RECOMMENDATIONS

1. Customer Retention

Protect the Largest Customer Segment

Develop loyalty programs and customer engagement initiatives focused on Cluster 3, which represents over half of the customer base.

2. Personalized Marketing

Segment-Based Campaigns

Use customer segmentation data to create tailored campaigns for different customer groups.

3. Revenue Growth

Upsell and Cross-Sell Strategy

Bundle Network & Infrastructure, Mobile Services, and Television & Entertainment packages together.

4. Product Portfolio Optimization

Focus on High-Performing Categories

Increase investment in:

- Network & Infrastructure
- Television & Entertainment
- Mobile Services

Review low-performing categories for improvement or repositioning.

5. Customer Experience Enhancement

Premium Service Programs

Provide proactive support and enhanced customer service for key customer segments.

Expected Impact :
↑ 15–25% retained revenue
Across all 5 strategic initiatives

08

FUTURE-STATE ROADMAP





PHASE 1: QUICK WINS

(0–3 MONTHS)

1



DEPLOY CUSTOMER SEGMENTATION DASHBOARD

Make data-driven decisions by visualizing customer segments and their performance.

CUSTOMER SEGMENTATION DASHBOARD

Total Customers

4.01K

Total Revenue

931.05M

Total Invoices

518K

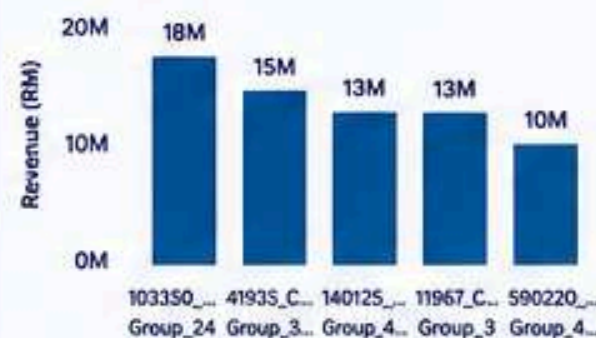
Avg. Revenue / Cust

232.18K

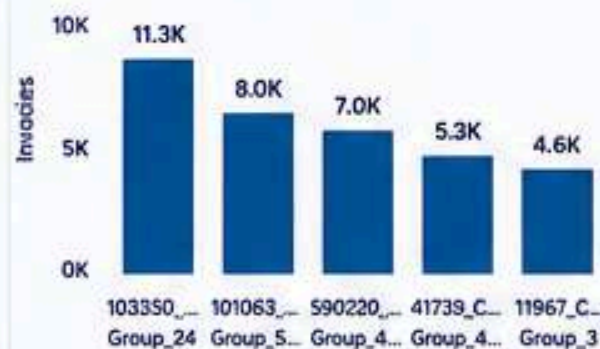
Customer Distribution by Cluster



Top 5 Customers by Revenue



Top 5 Customers by Invoice



Cluster Performance Snapshot

Cluster	Customers	Revenue (RM)	% Revenue
3	1,613	537.06M	57.70%
1	303	62.02M	6.66%
4	561	76.26M	8.19%
0	329	55.18M	5.93%
5	241	21.73M	2.33%
2	183	36.13M	3.88%
6	206	12.66M	1.36%
Total	4,010	931.05M	100.00%



Dashboard is live and accessible for stakeholders to explore customer segments and performance.

2



LAUNCH TARGETED MARKETING CAMPAIGNS

Reach the right customers with the right offers through the right channels.

SEGMENT-BASED CAMPAIGN STRATEGY



High Value Segment (Cluster 3)

1,613 Customers



Loyalty program,
exclusive offers &
priority support



Growth Segment (Cluster 1 & 4)

864 Customers



Cross-sell & bundle
offers to increase
engagement



At Risk Segment (Cluster 0, 2, 5, 6)

959 Customers



Retention offers,
personalized
incentives



Low Value Segment (Cluster 5 & 6)

447 Customers



Cost-effective
promotions &
awareness campaigns

CAMPAIGN CHANNELS



Email



SMS

Push
Notification

Social Media



Targeted campaigns are launched to drive engagement, conversion, and revenue growth.

3



MONITOR KEY SEGMENT PERFORMANCE

Continuously track segment performance to evaluate impact and refine strategies.

SEGMENT PERFORMANCE OVERVIEW



KEY MONITORING METRICS



Revenue

931.05M

↑+8.7%

vs. last 3 months



Invoices

518K

↑+9.3%

vs. last 3 months



Active Customers

4.01K

↑+6.2%

vs. last 3 months

Avg. Revenue/
Customer

Customer

232.18K

↑+2.4%

vs. last 3 months

PERFORMANCE INSIGHTS

- Cluster 3 continues to be the top revenue contributor (52.94%).
- Clusters 1 and 4 show steady growth in revenue and invoices.
- Clusters 0, 2, 5, and 6 require retention and engagement focus.



Regular monitoring ensures strategies are effective and resources are focused on the right segments.



PHASE 1 GOAL (0-3 MONTHS)

Establish a strong foundation by leveraging customer insights, engaging the right segments, and tracking performance to achieve quick wins.



Increase Engagement
& Conversion



Improve Customer
Satisfaction



Boost Revenue
Growth



Data-Driven
Decision Making



Phase 2: Business Optimization

(3–6 Months)

1



IMPLEMENT RECOMMENDATION ENGINES

Deliver the right products to the right customers

RECOMMENDATION ENGINE OVERVIEW

Recommendations
Generated

128.6K

Click-Through Rate
(CTR)

18.7% ↑

vs. baseline 12.3%

Conversion Rate

14.2% ↑

vs. baseline 9.1%

Incremental Revenue
(Pilot)

RM 6.45M

TOP RECOMMENDED PRODUCTS (PILOT)

Product Category

Recommendation Score



Mobile Services
Core

0.92



Broadband &
Internet Services
Core

0.87



Cloud & Hosting
Standard

0.85



Security Solutions
Standard

0.82



Digital Services
Standard

0.78

IMPACT (vs. NON-RECOMMENDED)



Cross-sell /
Up-sell Rate
+18%



Customer
Engagement
+21%



Revenue
Uplift
+15%



KEY BENEFITS

- ✓ Increase cross-sell and up-sell opportunities
- ✓ Improve customer satisfaction and engagement
- ✓ Drive incremental revenue through relevant recommendations

2



DEVELOP CHURN PREDICTION MODELS

Identify at-risk customers and take action early

CHURN RISK OVERVIEW

At Risk Customers

892
(22.2%)

Likely to Churn
(High Risk)

312
(7.8%)

Low Risk
Customers

2.45K
(61.0%)

Total Customers
Analyzed

4.01K

CHURN RISK DISTRIBUTION



TOP FACTORS INFLUENCING CHURN



MODEL PERFORMANCE (PILOT)

Accuracy

87.6%

Precision

85.1%

Recall

82.3%

AUC-ROC

0.90



KEY BENEFITS

- ✓ Detect at-risk customers early
- ✓ Reduce churn rate and revenue loss
- ✓ Improve customer retention and lifetime value

3



ENHANCE PRODUCT BUNDLING STRATEGIES

Increase value through smart product bundles

TOP PERFORMING PRODUCT BUNDLES (PILOT)

Bundle Name	Products Included	Customers	Revenue (RM)	Lift vs. Standalone
Digital Lifestyle Pack	📱 + 📶 + 📺	1,245	32.6M	+28%
Business Productivity Pack	☁ + 🛡 + 📱	982	27.4M	+24%
Home Entertainment Pack	📺 + 📶 + 📺	1,108	24.8M	+22%
SME Essentials Pack	☁ + 📱 + 👥	756	18.7M	+19%

BUNDLING IMPACT (vs. STANDALONE PRODUCTS)



Revenue
Uplift

+23%



Average Order
Value

+28%



Cross-sell
Rate

+31%



Customer
Adoption

38.6%

KEY BENEFITS

- ✓ Increase average order value and revenue
- ✓ Improve customer satisfaction with value-added bundles
- ✓ Strengthen customer loyalty and retention



OVERALL GOAL (3–6 MONTHS)

Leverage data and analytics to optimize customer experience, reduce churn, and maximize revenue through intelligent recommendations and product bundling strategies.



Improve Customer
Retention



Increase Revenue
& Profitability



Optimize Product
Portfolio



Drive Operational
Efficiency



PHASE 3: ADVANCED ANALYTICS

(6–12 MONTHS)

1



AI-DRIVEN CUSTOMER PERSONALIZATION

Deliver hyper-personalized experiences across the right products, channels and offers for each customer.

PERSONALIZATION IN ACTION



Customer ID
CUST_103350

Cluster 3
(High Value)

Top Recommended Products



Mobile Services

0.92



Broadband &
Internet Services

0.87



Cloud & Hosting

0.85

EXPECTED IMPACT



Higher Conversion
Rate

+15%

vs. baseline



Customer
Satisfaction

+20%

vs. baseline



Cross-sell /
Up-sell Rate

+18%

vs. baseline



Revenue
Uplift

+12%

vs. baseline

KEY BENEFITS

- ✓ Personalized offers based on customer behavior
- ✓ Right product, right time, right channel
- ✓ Stronger engagement and loyalty

2



REAL-TIME CUSTOMER ANALYTICS

Monitor customer behavior, performance and campaigns in real time to make faster and smarter decisions.

REAL-TIME ANALYTICS DASHBOARD

Active Customers Now

1,245



Today's Revenue (RM)

3.52M



Transactions Today

8,732



Campaign Performance (Today)



High 45%
Medium 35%
Low 20%

TOP EVENTS DETECTED (LIVE)



Spike in Mobile Services sign-ups
↑ 23% vs. yesterday

10:15 AM



High churn risk detected in Cluster 5
↑ 18% of customers at risk

10:32 AM



Broadband campaign performing well
CTR ↑ 14.8%

11:05 AM

KEY BENEFITS

- ✓ Real-time visibility into customer behavior
- ✓ Instant alerts and opportunity detection
- ✓ Faster response, better business outcomes

3



PREDICTIVE CUSTOMER LIFETIME VALUE (CLV) MODELING

Predict the total future value of each customer to prioritize retention and investment.

CLV SEGMENTATION OVERVIEW



TOP DRIVERS OF CLV



MODEL PERFORMANCE



R² Score **0.68**
MAE (RM) **215.4**
MAPE **18.7%**

KEY BENEFITS

- ✓ Identify and focus on high-value customers
- ✓ Improve retention and resource allocation
- ✓ Increase long-term customer profitability

4



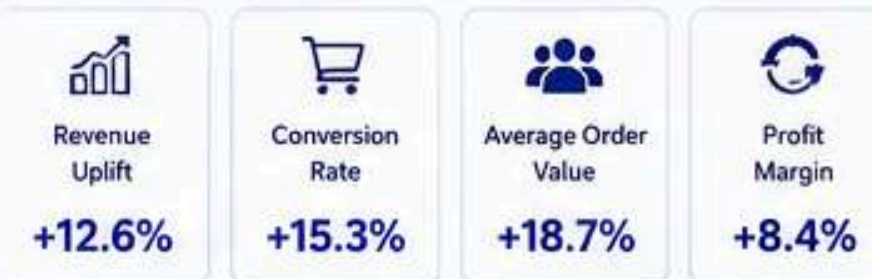
DYNAMIC PRICING & OFFER OPTIMIZATION

Use AI and analytics to optimize pricing and offers in real time to maximize revenue and customer value.

DYNAMIC PRICING SIMULATION



OFFER OPTIMIZATION IMPACT



PRICING BY PRODUCT CATEGORY (EXAMPLE)



KEY BENEFITS

- ✓ Real-time pricing maximizes revenue opportunities
- ✓ AI-optimized offers improve profitability
- ✓ Stay competitive with data-driven pricing



OVERALL GOAL (6-12 MONTHS)

Transform INFINI TELCO into a data-driven, AI-powered organization that maximizes customer lifetime value and ensures sustainable revenue growth.



Stronger Customer
Relationships



Higher Customer
Lifetime Value



Increased Revenue
& Profitability

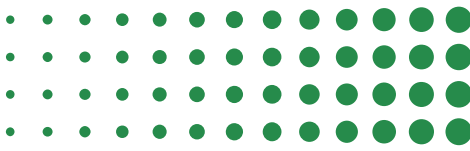


Data-Driven
Decision Making



Competitive
Advantage

»»» KEY FINDINGS



- The Powerhouse Segment (Cluster 3): Out of 8 identified behavioral clusters, Cluster 3 is the vital engine of Infini Telco. It comprises 1,613 customers who generate RM537.06 million (over 57% of total business revenue) across 364,280 invoices, maintaining the highest average revenue per customer at RM332,954.48.
- Portfolio Concentration: Total revenue reached RM931.05 million, heavily dominated by two top categories: Network & Infrastructure (RM151.5M) and Television & Entertainment (RM123.7M). Conversely, the bottom 5 categories combined (including Emerging Technologies and E-commerce Solutions) generate less than 3% of revenue.
- Product Type Adoption: Customers are highly concentrated around Core products, which represent 64.65% of customer distribution, while Standard products account for just 4%.





STRATEGIC RECOMMENDATION

- **Defend Cluster 3 via Key Account Management:** Implement premium service programs, proactive dedicated customer support, and tailored loyalty initiatives to insulate this high-value cluster from competitors.
 - **Bundle & Cross-Sell:** Package high-performing, high-adoption categories (Network & Infrastructure, Mobile Services, and Television & Entertainment) into attractive ecosystem bundles for high-activity, lower-value segments (e.g., Cluster 4).
 - **Product Portfolio Realignment:** Shift investment from stagnant, long-tail categories to scaling high-margin, top-performing core segments. Reposition underperforming lines like Data and Analytics (RM29M) via targeted commercial campaigns.
- 



INFINI TELCO



THANK YOU!



Thank you for your time, attention,
and support throughout our project on
Customer Segmentation
& **Product Category Analysis.**



DATA-DRIVEN INSIGHTS

Turning data into
meaningful business
decisions.



BETTER DECISIONS

Empowering smarter
strategies for growth
and efficiency.



STRONGER GROWTH

Driving value,
customer satisfaction,
and long-term success.



TOGETHER WE SUCCEED

Collaboration today,
sustainable impact
tomorrow.